

What should we expect before we simulate?

First-principles photon budget of a 1.4 m scintillator bar
for the SHiP Timing Detector

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Central idea

Geometry + Snell + exponential attenuation predict the dominant trends before any Geant4 output is shown. If simulation and napkin disagree, one of them is wrong — finding out which is the actual result.

The object: one bar of a 50 m² detector

Bar geometry: 1400 mm $\times$ 60.00 mm $\times$ 10.00 mm

- 8 SiPMs per END face
- Up to 70 SiPMs along TOP face
- ~ 600 bars for 50 m²

Best result tested so far

EJ-230 + END + TOP (20 SiPMs) + Vikuiti:

$\sigma_{\text{END}} = 53.68$ ps, $\sigma_{\text{TOP}} = 15.20$ ps, $\sigma_{\text{BLUE}} = 15.21$ ps

at $x = 0$, no electronic jitter.

Optimisation is still open

The best tested point is not proven to be the optimum.

The question is the Pareto front of $\sigma(T_0)$ vs. number of channels, with SiPM jitter and front-end jitter included.

1 — Napkin

Snell, geometry and exponential attenuation fix the order of magnitude and the *direction* of every trend: which material gives more light, which gives better timing.

2 — Geant4

Confirms the scale, adds distributions the napkin cannot produce: Landau fluctuations, path-length dispersion, angular mixing.

Knoll on optical Monte Carlo: these codes are accurate only to the extent that surface reflection, bulk absorption and refractive indices are correct. Those are exactly the inputs a napkin calculation can police.

Bethe–Bloch minimum, through $H = 10.00$ mm

$$\frac{dE}{dx} \approx 2.000 \text{ MeV/cm} \Rightarrow E_{\text{dep}} \approx 2.000 \text{ MeV}$$

- Sets the scale of everything downstream
- Every later factor multiplies this 2.000 MeV
- Width is Landau — simulation territory

What the napkin cannot give

Event-by-event Landau fluctuations, δ -ray production, path-length variation. The mean survives; the width is for Geant4.

$$N_{\gamma} = E_{\text{dep}} \cdot Y \approx 2 \times 9700 \approx 1.9 \times 10^4$$

Optical properties used in simulations:

Material	Yield [ph/MeV]	τ_d [ns]	λ_{att} [cm]
EJ-200	10000	2.1	380
EJ-204	10400	1.8	160
EJ-230	9700	1.5	120

The trade-off

Yields agree to within 7 %. What separates the materials is how much survives 70 cm travel (λ_{att}) and how fast the emission decays (τ_d) — and those two pull in opposite directions.

Critical angle

$$\theta_c = \arcsin(1/n) = 39.27 \text{ deg}, \quad n = 1.580$$

Trapped fraction — slab (Knoll Eq.8.15)

$$a = \cos \theta_c = 0.774, \quad f_{\text{TIR}} = 2a - 1 = 0.5484$$

Both walls: $f_{\text{TIR}}/\text{end} = 0.2742$

The bar is not a slab

Two wall families must hold TIR simultaneously.
Knoll Eq. 8.14 (cylinder, meridional rays) gives 18.4% per direction — different geometry.

Key number

≈ 0.2742 of photons are guided toward each end.

Group velocity

$$v_g = c/n \approx 189.7 \text{ mm/ns}$$

Effective velocity (END, from Geant4)

$$v_{\text{eff}} = 184.6 \text{ mm/ns} \Rightarrow \beta_{\text{eff}} \approx 13.37 \text{ deg}$$

β_{eff} is the effective axial angle of the photon population that dominates the earliest arrival times.

END vs. TOP select different photons

END is dominated by nearly axial, early photons. TOP integrates markedly more diagonal trajectories ($\beta_{\text{eff},\text{TOP}} \approx 40^\circ$). These two faces carry different timing information.

Geometric estimate, ray just outside TIR

$$N_{\text{bounce}} \approx \frac{d \tan \alpha_c}{D}, \quad \alpha_c = 50.73 \text{ deg}$$

With $d = 700 \text{ mm}$, $\tan \alpha_c = 1.223$:

Face	D	N_{bounce}
thin (H)	10.00 mm	≈ 86
wide (W)	60.00 mm	≈ 14

Factor of 6 difference

The same photon direction produces $\sim 6\times$ more bounces against the 10 mm face than the 60 mm face. Reflector quality matters more for the thin faces.

Step 6 — Reflector losses become exponential attenuation

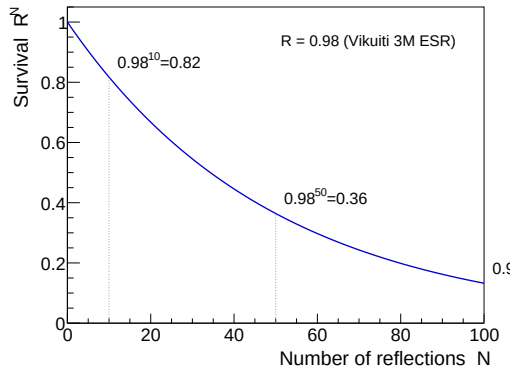
Survival after N bounces

$$P_R = R^N = e^{N \ln R}$$

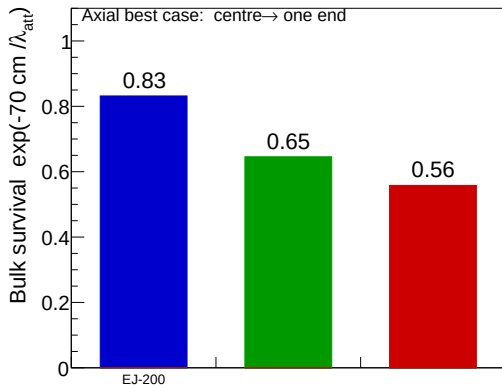
$$\Lambda_{\text{refl}} = \frac{-D}{\tan \alpha_c \cdot \ln R}$$

With $R = 0.98$ (Vikuiti, napkin):

Face	D	Λ_{refl}
thin (H)	10.00 mm	≈ 0.4 m
wide (W)	60.00 mm	≈ 2.4 m



Step 7 — Bulk attenuation and decay time pull in opposite directions



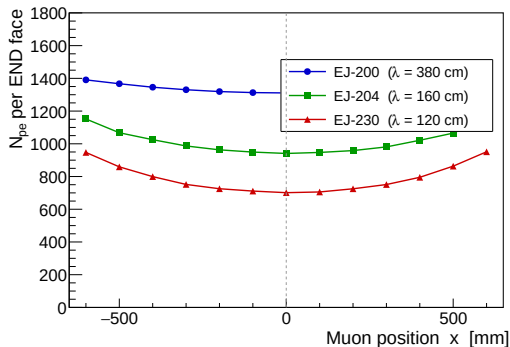
Napkin prediction (before the run)

EJ-200 delivers the most light (largest λ_{att}).
EJ-230 loses the most, but its faster kinetics ($\tau_d = 1.500 \text{ ns}$ vs. 2.100 ns) should win on timing.

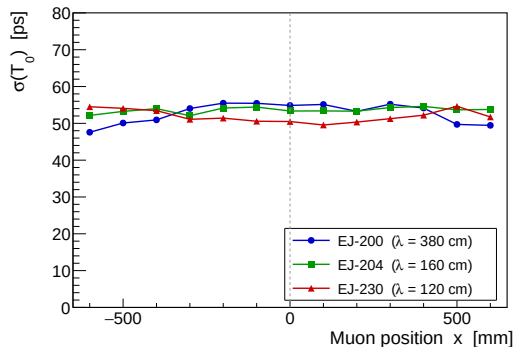
Geant4 confirms both orderings

N_{pe} : EJ-200 > EJ-204 > EJ-230
 $\sigma(T_0)$: EJ-230 < EJ-204 < EJ-200
Both directions as predicted.

The ordering was predicted, then observed



N_{pe} per END vs. position. EJ-200 > EJ-204 > EJ-230, steeper fall for shorter λ_{att} .



$\sigma(T_0)$ per END vs. position. Ordering inverted: EJ-230 < EJ-204 < EJ-200, driven by τ_d .

Two figures, two orderings, opposite directions — written down before the simulation ran.

Closing the chain: photon budget for EJ-230 at $x = 0$

Step	Factor	Cumulative photons
Produced	$Y \times E_{\text{dep}} = 9700 \times 2$	$\sim 1.94 \times 10^4$
TIR-guided toward one end	$\times 0.2742$	~ 5300
Bulk survival (70 cm)	$\times 0.5580$	~ 2970
End-face coverage (288/600)	$\times 0.4800$	~ 1430
PDE (≈ 0.4000)	$\times 0.4000$	$\sim 570.0 \text{ pe}$
Geant4 (EJ-230, END, Vikuiti, $x = 0$)	—	$\approx 1400 \text{ PE}$

Scale check, not a prediction

The chain closes to within 20 %. The excess is from the reflector-recovered population. Factors are not strictly independent.

Count per end — always

Mixing a both-ends fraction with a one-end coverage silently doubles the answer.

Readout coupling	Acceptance toward one end	PE/end	Vik/TIR ratio
Index-matched	0.2742 (TIR-guided)	≈ 570.0 pe	≈ 1.2
Air gap (escape cone)	0.1129	≈ 234.6 pe	≈ 3
Reported run	—	≈ 1 PE	$\approx 10^3$

Escape cone cannot explain 10^3

The cone cuts both Vikuiti and TIR configurations by the same factor at the readout face — it cannot generate a ratio. The napkin brackets the genuine ratio between 1 and 3.

Falsifiable prediction

Run END + TIR-only and END + Vikuiti under identical conditions. A ratio near 10^3 is a surface or RINDEX definition to be found, not a physics result.

Two-end readout

$$t_L = t_0 + (L/2 + x)/v_{\text{eff}} + \varepsilon_L$$

$$t_R = t_0 + (L/2 - x)/v_{\text{eff}} + \varepsilon_R$$

Sum: $T_0 = (t_L + t_R)/2$, x cancels

Diff: $\Delta t = -2x/v_{\text{eff}} \Rightarrow$ position

$$\sigma_x = (v_{\text{eff}}/2) \cdot \sigma_{\Delta t} \approx 10.29 \text{ mm}$$

Consistency check

$v_{\text{eff}} = 184.6 \text{ mm/ns}$, $\sigma_{\Delta t} = 111.5 \text{ ps}$:

$$\sigma_x \approx 10.29 \text{ mm.}$$

Position resolution is not independent information
— it is derived from the timing resolution.

Path-length spread limits timing

For large scintillators and fast emitters, the distribution of t_{prop} is the dominant limit (Knoll).

Quantity	Napkin / literature	Verdict
Photons per muon	$\approx 2 \times 10^4$	consistent
Trapped fraction (one wall)	$a = 0.774$, Knoll 8.15	textbook match
Group velocity	≤ 189.7 mm/ns	consistent
N_{pe} ordering	EJ-200 > EJ-204 > EJ-230	confirmed
$\sigma(T_0)$ ordering	EJ-230 < EJ-204 < EJ-200	confirmed
σ_x from $\sigma_{\Delta t}$	10.29 mm	internal check
Cylinder trapping (EJ-228)	18.4%/dir (Eq. 8.14)	different defn.
Vikuiti/TIR yield ratio	between 1 and 3	audit the run

6 agreements. 1 needing definition check. 1 genuine anomaly.

Two photon populations carry different information

Population A — TIR-guided

Large $|u_x|$, few bounces, v_x close to c/n , arrives first. The leading-edge (earliest- k) estimator is dominated by this population.

Population B — reflector-recovered

Large transverse component, many bounces, R^N penalty, arrives late. Vikuiti raises total N_{pe} mainly through this population.

Key message

Total $N_{pe} \neq$ timing information carried by the *first* photons.
A reflector that doubles N_{pe} does not halve $\sigma(T_0)$.

Specular — Vikuiti 3M ESR

Preserves angle of incidence. Keeps path-length dispersion small, favouring the earliest photons.

Diffuse — Teflon, TiO_2 , MgO

Lambert's law: outgoing direction independent of incoming. Every bounce is a fresh chance to redirect toward a readout window, at the cost of a broader t_{prop} distribution.

Napkin hypothesis (falsifiable)

At comparable reflectivity, diffuse scattering should raise collected light and broaden the arrival distribution. One campaign, one variable, prediction on record.

Knoll: diffuse reflectors generally collect more light, but that comparison is on yield, not timing.

Why assume the first photon is optimal?

Current estimators:

- Leading-edge: use $t^{(k)}$ for $k = 1$
- Trimmed mean of first m photons
- BLUE combination of END + TOP

Evidence that $k = 1$ is not always optimal

In an EJ-228 cylinder study, $\sigma(T_0)(k)$ is non-monotonic with minimum near $k \approx 8$. The number does not transfer; the principle does.

Best tested result (EJ-230, $x = 0$)

$\sigma_{\text{END}} = 53.68 \text{ ps}$ (END, $m^* = 8$)

$\sigma_{\text{TOP}} = 15.20 \text{ ps}$ (TOP, $k^* = 3$)

$\sigma_{\text{BLUE}} = 15.21 \text{ ps}$ (BLUE combination)

This is the optical limit only. If SiPM SPTR $\sim 100 \text{ ps}$, winning 14 ps optically matters less than halving the channel count.

- A — **Exploit existing data** Audit ROOT files for arrival time, SiPM ID, face, PE and true muon position. Fix the per-end vs. L+R convention before any ratio is quoted.
- B — **Close the 1 PE question** Run END + TIR-only and END + Vikuiti under identical conditions. Napkin brackets the per-end yield ratio between 1 and 3; anything near 10^3 is a surface or RINDEX definition problem.
- C — **Separate online/offline ablation** Masking a channel offline $\neq$ removing it from the geometry: physical removal changes the optical boundary condition.
- D — **Report a Pareto front, not a minimum** $\sigma(T_0)$ (mean and worst case in x) vs. number of channels, with SPTR and front-end jitter injected.

1 — Geometry explains the scale

TIR, propagation and repeated reflection predict the dominant trends and reproduce the textbook where the textbook applies.

2 — More light is not necessarily better timing

Early TIR photons and late reflector-recovered photons carry different information. Total N_{pe} is the wrong figure of merit for a timing detector.

3 — The optimisation is still open

The optimum is not the maximum number of SiPMs. It is the Pareto optimum of $\sigma(T_0)$ vs. channel count, once SPTR and front-end jitter are in.

EJ-230 + END + Vikuiti is the best *tested* point — not yet the optimum.